

SX906

Next-Gen SmartNIC

Powered by Intel® Xeon® 6 SoC



SX906 is based on Intel® Xeon® 6 SoC processor to offload x86 applications with zero changes. SX906 delivers high-performance compute in a compact PCIe form factor, reducing infrastructure footprint while ensuring rapid service deployment.

- Applications:
- SD-WAN/SD-LAN
 - Next-Gen Firewall (NGFW)
 - Unified Threat Management (UTM)
 - Zero-Trust Network Access (ZTNA)
 - AI Edge and Cloud-Native Security Services

Key Features

- Intel® Xeon® 6 SoC Processor
- Intel® Ethernet Controller E830 for 200GbE PCIe connection to host CPU
- Dual 100G QSFP28
- Four DDR5 memory channels
- TPM2.0 support for Secure Boot
- OpenBMC, Intel® Platform Firmware Resilience (Intel® PFR)
- Dual Slot Active Cooling
- MCIO Connectors (Gen5 x8) for Function Expansion

Physical

- Form Factor: Dual Slot, FH10.5L(266mm)
- Dimensions: W266xH98.4xD40.6mm
- Weight: 1.0kg

Power

Power delivered through PCIe edge finger and 16-pin PCIe AUX Power connector (12VHPWR)

I/O Configuration

- 2 x QSFP28
- 1 x MiniDP
- 1 x USB3.0 Type-C
- 1 x USB3.0 Type-C Console
- 1 x RJ45 1GbE Management

Key Components

- Intel® Xeon® 6 Granite Rapids-D
- Intel® Ethernet Controller E830
- Aspeed AST2600/ AST1060

Environmental

- Operating Temperature: 0°C to 45°C
- Storage Temperature: -40°C to 70°C
- Operating Humidity: up to 85% non-condensing

OS

- Ubuntu 22.04/ 24.04

Processor	CPU Cores	Base Frequency	Processor Total Bandwidth	Power Consumption	Networking	Host Interface	Memory Support	Storage Capacity	BMC FPR
Xeon® 6523P-B	24C	2.5GHz	100G	295	2x100GbE QSFP28	PCIe Gen5 x8	Up to 64GB, 4-channel DDR5-4600 ECC	Up to 128GB eMMC	AST1060 (Intel® PFR)
Xeon® 6553P-B	36C	2.6GHz	200G	355	2x100GbE QSFP28				AST2600 (OpenBMC/BIOS)
Xeon® 6563P-B	38C	2.4GHz	200G	355	2x100GbE QSFP28				

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I/O Interfaces

